Session 3: Stellar Structure Basics

Hydrostatic equilibrium, polytropes, the Lane–Emden equation

astro-lab – reading notes

Goals

Derive the equations of a simple stellar model, reduce them to one dimensionless ODE, and solve it numerically for several polytropic indices, checking against analytic solutions.

1 Equations of structure

For a spherical star in equilibrium, with $m(r)$ the mass inside radius r:

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \rho, \tag{1}$$

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{G\,m\,\rho}{r^2}. \tag{2}$$

Equation (2) (hydrostatic equilibrium) says pressure gradient balances gravity. A real star also needs energy transport and an equation of state; a polytrope sidesteps them with a single assumption.

2 Polytropes

Assume $P = K\rho^{1+1/n}$, where n is the polytropic index. Rewrite (2) as $\frac{1}{\rho}\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm}{r^2}$, multiply by r^2 and differentiate, then use (1):

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{r^2}{\rho}\frac{\mathrm{d}P}{\mathrm{d}r}\right) = -4\pi G\rho.$$

Write $\rho = \rho_c \theta^n$ and $r = a\xi$ with

$$a^2 = \frac{(n+1)K\rho_c^{1/n-1}}{4\pi G}.$$

Substituting gives the **Lane–Emden equation**

$$\frac{1}{\xi^2}\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\right) = -\theta^n, \qquad \theta(0) = 1, \quad \theta'(0) = 0.$$

The surface is the first zero ξ_1 where $\theta(\xi_1) = 0$.

Behaviour near the centre

The equation is singular at $\xi = 0$. A series gives

$$\theta \approx 1 - \frac{\xi^2}{6} + \frac{n\,\xi^4}{120} - \cdots$$

so a numerical solver should start at a small $\xi_0 > 0$ using the series values, which is exactly what the script does.

3 Analytic checks

n	Solution $\theta(\xi)$	ξ_1
0	$1 - \xi^2/6$	$\sqrt{6} \approx 2.4495$
1	$\sin\xi/\xi$	$\pi \approx 3.1416$
5	$(1 + \xi^2/3)^{-1/2}$	∞ (infinite radius, finite mass)

Numerical values from the repository script, which agree with the standard tabulated values:

n	ξ_1
0	2.44949
1	3.14159
1.5	3.65375
3	6.89685
4	14.97155

$n = 1.5$ models a fully convective star or a non-relativistic degenerate gas (white dwarf); $n = 3$ is Eddington's standard model for radiation-dominated, high-mass stars.

4 Mass and central density

Integrating (1) with $\rho = \rho_c \theta^n$ gives

$$M = 4\pi a^3 \rho_c \left(-\xi^2 \frac{\mathrm{d}\theta}{\mathrm{d}\xi} \right)_{\xi_1}, \qquad R = a\,\xi_1.$$

The mean-to-central density ratio is $\bar{\rho}/\rho_c = -\dfrac{3}{\xi_1}\theta'(\xi_1)$. For $n = 3$ this is about $1/54$, so such stars are strongly centrally concentrated.

5 Code walkthrough

```
"""Session 3: Lane-Emden equation  (1/x^2) d/dx (x^2 dθ/dx) = -θ^n.
Checks: n=0, 1, 5 have analytic solutions. Prints first zero ξ1 (stellar surface).
"""
import numpy as np
from scipy.integrate import solve_ivp


def solve(n, x_max=50.0):
    def rhs(x, y):
        th, dth = y
        return [dth, -max(th, 0.0) ** n - 2.0 * dth / x]

    x0 = 1e-6
    th0 = 1 - x0**2 / 6  # series start avoids the x=0 singularity
```

```
    dth0 = -x0 / 3
    hit = lambda x, y: y[0]
    hit.terminal, hit.direction = True, -1
    sol = solve_ivp(rhs, [x0, x_max], [th0, dth0], events=hit, rtol=1e-10, atol=1
        e-12, dense_output=True)
    return sol


if __name__ == "__main__":
    analytic = {0: np.sqrt(6), 1: np.pi}
    for n in (0, 1, 1.5, 3, 4):
        xi1 = solve(n).t_events[0][0]
        ref = f"  (analytic {analytic[n]:.5f})" if n in analytic else ""
        print(f"n={n}: xi1={xi1:.5f}{ref}")
    # known: n=1.5 -> 3.65375, n=3 -> 6.89685 (Eddington standard model)
```

Notes on the implementation: the right-hand side uses $\max(\theta, 0)^n$ so the solver does not take a fractional power of a small negative number; the terminal event $\theta = 0$ stops the integration at the surface.

6 Exercises

1. Plot $\theta(\xi)$ for $n = 0, 1, 1.5, 3, 4$ on one axes and compare with the analytic curves where they exist.
2. Compute M and R for an $n = 3$ polytrope with the Sun's M and R; solve for K and ρ_c. Compare ρ_c with the true solar core density ($\sim 150\ \mathrm{g\,cm^{-3}}$).
3. Why does $n = 5$ have no finite surface? Show $\theta = (1 + \xi^2/3)^{-1/2}$ satisfies the equation.
4. Build a Gaia HR diagram (Session 1) and mark where $n = 1.5$ and $n = 3$ models would sit qualitatively.

References

- Hansen, Kawaler, Trimble, *Stellar Interiors*, ch. 7
- Kippenhahn, Weigert, Weiss, *Stellar Structure and Evolution*
- Chandrasekhar, *An Introduction to the Study of Stellar Structure*, ch. IV